

Shield EPC Verification and Evidence Framework

1. Purpose

This framework ensures that every change to the Shield EPC repository — from a routine fix to an architecture decision — is backed by verifiable, inspectable evidence rather than assumption or intent.

It formalizes the evidence-first discipline: negative claims are not accepted without a source check, and status changes are not accepted without reproducible evidence tied to a specific commit.

2. Scope

Applies to all code changes, ADR status transitions, domain model changes, and architecture-affecting decisions in Shield EPC. It does not replace code review; it defines the minimum evidence record required before a change or claim can be treated as verified.

3. Verification Principles

- **Evidence over intent.** An ADR or status is promoted based on verified evidence, not architectural intent alone.
- **No negative claims without inspection.** A statement that something "does not exist" or "is not implemented" must cite the source location inspected to confirm its absence.
- **Reproducibility.** Every verification is tied to a specific commit hash so it remains a valid historical statement even after the codebase or test suite changes.
- **Proportionality.** Evidence depth scales with risk and blast radius, not with change size. A one-line fix can require full verification if it touches a domain boundary; a large but mechanical refactor may not.

4. Verification Levels

Level	When to use
Routine	Minor, low-risk changes (typos, styling, isolated logic tweaks) with no architectural surface
Short	Feature completion or moderate changes where structural evidence matters but full architectural review is unwarranted

Level	When to use
Full	ADR status changes to Verified Stable or Frozen, domain model changes, or anything hitting the Escalation Rule

5. Escalation Rule

If a Routine or Short Verification surfaces an unexpected architectural finding, a negative-claim contradiction, an ADR impact, a domain model change, or a public API/database behavior change, the verification is immediately escalated to a Full Verification Record — regardless of how small the original change appeared.

Evidence takes precedence over the original change's perceived size.

6. Short Verification Record (Template)

```
### Short Verification Record

**Feature / Change:**
**Commit Hash:**
**Date:**

#### Source Inspection
```

```
- Files inspected:
- File:Line references:
- Expected vs Observed:
```

```
#### Targeted Tests
```

```
- Tests executed:
- Result:
```

```
#### Conclusion:
```

7. Full Verification Record (Template)

```
### Full Verification Record
```

```
**Feature / Change:**
**Status Change:**
**Date:**
```

```
#### Evidence Baseline
```

```
- Repository:
- Commit Hash:
- Version Tag:
- Branch:
```

```
#### Verification Method
```

```
**1. Source Inspection**
```

```
- Files inspected:
- File:Line references:
- Expected implementation:
```

- Observed implementation:

****2. Negative Claim Verification****

- Claim being verified:
- Evidence inspected:
- Result:

****3. Targeted Tests****

- Tests executed:
- Result:

****4. Full Regression****

- Test Suite Snapshot:
 - Passed:
 - Failed:
 - Skipped:
- Commit Hash:
- Environment (OS, runtime/interpreter version, dependency lock file hash):

Impact Assessment

- ADR affected:
- Domain model affected:
- Public API affected:
- Database schema affected:
- Breaking changes:

Evidence Quality

- Direct evidence:
- Indirect evidence:
- Assumptions:
- Confidence Level: High / Medium / Low
(governed by weakest link, not the average)

Verification Metadata

- Verified By:
- Review Date:
- Review Notes:

Evidence Summary

- Conclusion:
- Remaining work:
- Follow-up required:

8. Evidence Quality Guidance

- **Direct evidence:** Something the reviewer personally read in source code, ran in a test, or observed in runtime output. This is the only evidence that can close a negative claim.
- **Indirect evidence:** Inference from related code, documentation, or a teammate's report. Usable, but must be labeled as indirect and should not by itself justify promotion to Verified Stable or Frozen.
- **Assumptions:** Anything taken on faith because full inspection was impractical. Must be listed explicitly, never left implicit — an unstated assumption is how "verified" claims silently rot.
- **Confidence Level** reflects the weakest link in the evidence chain, not the average. One unverified assumption caps the whole record at Medium,

regardless of how much direct evidence surrounds it.

- **Runtime output must be tied to the commit under verification.** A log, test run, or console output from a different commit is not valid evidence for the current record, even if the code "looks the same" — re-run it against the exact commit hash being verified.

9. Usage Examples

Example — Routine:

A one-line null-check fix in a utility function. Commit hash + the one affected test + pass count is sufficient; no template required.

Example — Short:

A new read-only API endpoint added to an existing, unaffected module. Use the Short Verification Record — source inspection plus targeted tests covering the new endpoint.

Example — Full:

An ADR status change from Verified Stable to Frozen for a major architectural component, closing a previously identified gap. Requires the Full Verification Record, including the full regression snapshot and impact assessment, per the Escalation Rule.

10. Governance Notes

- This template supersedes ad hoc verification notes in docs/ADR_INDEX.md going forward; existing entries are not required to be retrofitted unless they undergo a future status change.
- **AI-assisted verification** (including AI-authored source inspection or test summaries) is subject to the same evidence-first rule as human-authored claims: a tool's statement that an artifact was created or a check was performed is not itself evidence. The artifact must be independently opened and inspected before the claim is accepted. AI tools may assist in drafting, searching, or summarizing evidence, but cannot substitute for it — the record's evidence fields must point to something a human independently inspected.
- This document should be referenced, not duplicated, from docs/ADR_INDEX.md and other architecture documents.